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Formula Sheet

Machine Learning I – SoSe 2025

1 Unsupervised Learning

1.1 Clustering

1.1.1 K-Means Clustering

Minimise total within cluster distance

$$W(C_1, \dots C_k) = \sum_{k=1}^{K} W(C_k) = \sum_{k=1}^{K} \sum_{i \in C_k} \|\boldsymbol{x_i} - \boldsymbol{\mu_k}\|^2, \text{ where } \boldsymbol{\mu_k} = (\mu_{k,1}, \dots \mu_{k,p}),$$

with

$$\mu_{k,j} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{i,j}$$

and $\|\boldsymbol{x} - \boldsymbol{y}\|^2 = \sum_{j=1}^{p} (x_j - y_j)^2$ being the squared *euclidean distance* between vectors $\boldsymbol{x}$ and $\boldsymbol{y}$.

For K-Medians use the Manhattan distance: $\|\boldsymbol{x} - \boldsymbol{y}\| = \sum_{j=1}^{p} |x_j - y_j|$.

2 Supervised Learning

2.1 Regression

2.1.1 Mean Square Error (MSE)

$$\text{MSE}(f, \boldsymbol{x}, \boldsymbol{y}) = \frac{1}{n}\sum_{i=1}^{n}(y_i - f(\boldsymbol{x}_i))^2 = \frac{1}{n}RSS \text{ (Residual sum of squares)}$$

2.1.2 Linear Models

A linear model has the form

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + ... + \beta_p x_{ip} + \epsilon_i y,$$

which in vector and matrix form is $\boldsymbol{y} = \boldsymbol{X}\beta + \epsilon$. $\boldsymbol{X}$ is called the design matrix and β is the vector of parameters.

The linear model estimates are the least squares estimates $\widehat{\beta}$, which minimises the residual sum of squares leading to $\widehat{\beta} = \boldsymbol{(X'X)^{-1}X'y}$.

- Fitted values: $\widehat{\boldsymbol{y}} = \boldsymbol{X}\widehat{\beta}$
- Residuals: $\widehat{\boldsymbol{\epsilon}} = \boldsymbol{y} - \widehat{\boldsymbol{y}}$

An **Anova model** is a linear model where the explanatory variables are discrete or nominal. Example one factorial Anova model, an overall parameter β_0 is fitted and then one parameter is fitted for all but the first level.

$$y_i = \beta_0 + \beta_1\mathcal{I}(x_i \in A_2) + \cdots + \beta_{K-1}\mathcal{I}(x_i \in A_K) + \epsilon_i$$

$\mathcal{I}(x_i \in A_k)$ is equal to 1 if x_i is in the group A_k and zero if not. A_1 is called the base line group.

2.1.3 Ridge Regression

Ridge regression minimises

$$RSS + \lambda\sum_{j=1}^{p}\widehat{\beta}_j^2 = \sum_{i=1}^{n}\left(y_i - \widehat{\beta}_0 - \sum_{j=1}^{p}\widehat{\beta}_j x_i\right)^2 + \lambda\sum_{j=1}^{p}\widehat{\beta}_j^2$$

The Lasso minimises

$$RSS + \lambda\sum_{j=1}^{p}|\widehat{\beta}_j| = \sum_{i=1}^{n}\left(y_i - \widehat{\beta}_0 - \sum_{j=1}^{p}\widehat{\beta}_j x_i\right)^2 + \lambda\sum_{j=1}^{p}|\widehat{\beta}_j|$$

2.2 Binary classification

Predict $Y=0$ if $P(Y=1|x_1, \ldots, x_p) < \alpha$ else predict $Y=1$.

2.2.1 Logistic regression

Logit function: $\text{logit}(s) = \log\left(\frac{s}{1-s}\right)$

Logistic regression linear predictor $\text{logit}(f(x_1, \ldots, x_p)) = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p$

$$\Rightarrow f(x_1, \ldots, x_p) = P(Y=1|x_1, \ldots, x_p) = \frac{\exp\{\beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p\}}{1 + \exp\{\beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p\}}$$

2.2.2 Model evaluation

- The **sensitivity** of a classifier is the true positive rate.
- The **specificity** of a classifier is the true negative rate.
- Important: ROC curve and AUC.

Bayes Theorem: The conditional probability of A given that B occurs is $P(A|B) = \frac{P(A \cap B)}{P(B)}$ which can be derived from $P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\bar{A})P(\bar{A})}$.

2.2.3 Bayes Classifier

Choose the group y_j which maximises $P(Y = y_k|x_1, \ldots x_p)$, for all $k = 1, \ldots K$

2.2.3.1 Linear discriminant analysis (LDA)

The LDA classifier rule is to choose the value of k giving the largest $P(Y = y_k|x)$, the posterior probability our outcome variable is in group y_k given x

$$P(Y = y_k|x) = \frac{f_{X|Y=y_k}(x)P(Y = y_k)}{\sum_{j=1}^{K} f_{X|Y=y_j}(x)P(Y = y_j)}$$

$P(Y = y_k)$ is the prior probability that our outcome variable is in group y_k.

The density $f_{X|Y=y_k}(x)$ in each group is modelled using a multinomial distribution:

$$(X_1, X_2, \ldots X_p)\,|\,Y = y_k \sim N(\boldsymbol{\mu_k}, \Sigma^2)$$

$\boldsymbol{\mu_k}$ is a vector of length p. Σ^2 is the common variance-covariance matrix.

2.2.3.2 Quadratic discriminant analysis (QDA)

As LDA but use a different covariance in each group.

$$(X_1, X_2, \ldots X_p)\,|\,Y = y_k \sim N(\boldsymbol{\mu_k}, \Sigma_k^2)$$

Σ_k^2 is the variance-covariance matrix in group k.

2.3 Tree Methods

2.3.1 Regression Trees

Minimise $RSS = \sum_{j\in|T|}\sum_{i\in R_j}(y_i - \widehat{y}_j)^2$, where the R_js are terminal nodes in the Tree T and $\widehat{y}_j = \frac{1}{|R_j|}\sum_{i\in R_j} y_j$ mean of Y in the region R_j.

Pruning The full tree T_0 is be pruned back to the subtree $T_\alpha \subset T_0$ by minimising

$(PLS(\alpha) = \sum_{j\in|T|}\sum_{i\in R_j}(y_i - \widehat{y}_j)^2 + \alpha|T|$, with complexity parameter $\alpha \geq 0$, and each R_j the terminal nodes in T.

2.3.2 Classification trees

- Classification error rate:
  - Single Node m: $E_m = \sum_{m=1}^{M}(1 - \max_k\{\widehat{p}_{mk}\})$
  - Complete Tree: weighted mean of all $E_{terminal\ nodes}$

$$E_{tree} = \frac{1}{N}\sum_{m=1}^{M}|T_m|\left(1 - \max_k\{\widehat{p}_{mk}\}\right)$$

- Gini Index:
  - Node: $G_{node} = \sum_{k=1}^{K}\widehat{p}_{mk}(1 - \widehat{p}_{mk})$
  - Tree: weighted mean of all $G_{terminal\ nodes}$

$$G_{tree} = \frac{1}{N}\sum_{m=1}^{M}|T_m|\sum_{k=1}^{K}\widehat{p}_{mk}(1 - \widehat{p}_{mk})$$

- Entropy or Information index:
  - Node: $D_m = -\sum_{k=1}^{K}\widehat{p}_k\log(\widehat{p}_k)$
  - Tree: weighted mean of all $D_{terminal\ nodes}$

$$D_{tree} = -\frac{1}{N}\sum_{m=1}^{M}|T_m|\sum_{k=1}^{K}\widehat{p}_{mk}\log(\widehat{p}_{mk})$$